Use Case ID:	Use Case 1		
Use Case Name:	Exploring user profile page		
Created By:	Li Shuhui	Last Updated By:	Li Shuhui
Date Created:	2025.9.15	Date Last Updated:	2025.9.18

Actor:	-User (Student, Young Working Adult, Household) -System
Description:	Allow user to manage account settings, meal plans, grocery lists, and personalization preferences.
Preconditions:	-User has internet access. -User has registered an account and already login. -Login page is accessible.
Postconditions:	The user leaves the current page or switches to another module of the system.
Priority:	Medium
Frequency of Use:	Occasionally
Flow of Events:	1.User opens Profile Page. 2.System displays name, avatar, saved meal/grocery lists, and defaults.
Alternative Flows:	1.If user edits meal plans or grocery list -System auto-save the current plan draft and show “Last saved” timestamp. -System updates data and shows confirmation. 2.If user toggles dark/light mode - The system changes the mode accordingly. 3.If user clicks “Data Sources” - Then system display CPI/NEA/price datasets.
Exceptions:	1.If system response > 2s -System displays a loading message until the chart is ready.
Includes:	All the actions include login.
Special Requirements:	Price values must be within ± 0.05 SGD of the dataset.
Assumptions:	-Dataset is always available and up-to-date. -Users have stable internet access to view dynamic charts.
Notes and Issues:	-If many meal/grocery lists are stored the interface may be very long and also need large storage.

Use Case ID:	Use Case 2		
Use Case Name:	View Food Price Trends		
Created By:	Li Shuhui	Last Updated By:	Li Shuhui
Date Created:	2025.9.18	Date Last Updated:	2025.9.18

Actor:	-User (Student, Young Working Adult, Household) -System
Description:	The Food Price Inflation Dashboard enables users to track changes in food prices over time.
Preconditions:	-CPI dataset is available and up to date.
Postconditions:	The user leaves the current page or switches to another module of the system.
Priority:	Medium
Frequency of Use:	Occasionally
Flow of Events:	1.User navigates to the Food Price Inflation Dashboard. 2.User selects food items or categories to be inquired. 3.System retrieves CPI (Consumer Price Index) data for selected choices. 4.User selects a time range (e.g., last 6 months). 5.System generates a line chart for monthly prices. 6.System displays numerical price values in SGD (2 decimals).
Alternative Flows:	1.If user modifies the filters -System asks for selecting a time range. -System returns to step 5 2.If user adjusts the time range - System updates the chart accordingly. 3.If user choose to type in keywords for specific food - System shows the items according to the keywords for user to choose. - System returns to step 3.
Exceptions:	1.If CPI data is missing for some months -System shows “Data unavailable” -System continues with available months. 2.If system response > 2s -System displays a loading message until the chart is ready.
Includes:	-Generating line charts includes showing exact numerical prices. -Retriving CPI food data includes selecting a time range.

	-Searching food items by keywords includes adding items into comparison. -Retriving CPI food data includes Generating line charts.
Special Requirements:	Price values must be within ± 0.05 SGD of CPI dataset.
Assumptions:	-CPI dataset is always available and up-to-date. -Users have stable internet access to view dynamic charts.
Notes and Issues:	-Historical data gaps may cause incomplete charts. -Large time ranges may increase response time slightly

Use Case ID:	Use Case 3		
Use Case Name:	Budget Meal Planner		
Created By:	Li Shuhui	Last Updated By:	Li Shuhui
Date Created:	2025.9.15	Date Last Updated:	2025.9.18

Actor:	-User (Student, Young Working Adult, Household) -System
Description:	The Budget Meal Planner helps users design affordable daily or weekly meal plans.
Preconditions:	-User has internet access. -User has logged in.
Postconditions:	The user leaves the current page or switches to another module of the system.
Priority:	High
Frequency of Use:	Frequently
Flow of Events:	1.User navigates to the Budget Meal Planner. 2.User sets daily/weekly budget in SGD. 3.User selects dietary rules (e.g., halal, vegetarian). 4.User selects calorie range. 5.User chooses “Eat out” or “Cook at home”. 6.User sets the number of meals per day and the plan horizon.

	6.System generates 2-3 meal plans with itemized costs, food options, total cost and nutrition breakdown.
Alternative Flows:	1.If user clicks “Browse” - The system provides cheaper or alternative options that still fit the meal rules. -System shows estimated savings. 2.If user saves plan -System stores the plan under the user’s ID. 3.If System provide a plan exceeding the budget or the saved plans exceeding the current budget after the user changes the budget -System sets up an alert for the user. -System provides alternative plans to keep total cost under budget if possible.
Exceptions:	1.If no valid meal plan fits budget -System suggests closest alternatives and highlights cost overrun. 2.If system response > 5s -System displays a loading message until the plan is ready.
Includes:	-Setting a budget includes selecting a time range. -Selecting meal rules includes setting the number of meals per day. -Generating meal plan includes providing nutritional breakdown and displaying total cost.
Special Requirements:	Price estimates must be within ±10% of CPI-adjusted baseline.
Assumptions:	-Users provide reasonable dietary and calorie inputs. -Food pricing data is current and covers common meal ingredients.
Notes and Issues:	-Substitutions must always include a clear explanation for transparency. -Some dietary restrictions (e.g., vegan + halal combined) may limit available meal options.

Use Case ID:	Use Case 4		
Use Case Name:	Find Cheaper Hawker Alternatives		
Created By:	Li Shuhui	Last Updated By:	Li Shuhui

Date Created:	2025.9.15	Date Last Updated:	2025.9.19

Actor:	-User (Student, Young Working Adult, Household) -System
Description:	The Smart Hawker Swap suggests cheaper hawker food alternatives to help users cope with rising dish prices.
Preconditions:	-User has logged into the website. -User has allowed location access .
Postconditions:	-User leaves the current page or switches to another module of the system.
Priority:	High
Frequency of Use:	Frequently
Flow of Events:	1.User navigates to Smart Hawker Swap. 2.System displays average dish prices nearby from CPI/NEA datasets. 3.System displays food categories with high inflation. 4.User searches for a dish through a search bar. 5.System shows location details and directions to stalls. 6.System recommends cheaper alternatives within 3 km radius if possible. 7.System displays estimated savings.
Alternative Flows:	1.If user selects a cuisine filter -System provides suggestions by cuisine type. -System returns to step 5. 2.If user requests route map -System opens location in map view. 3. If user choose to enter the address -System uses the entered address instead of the current address when using address information.
Exceptions:	1.If no hawker stalls found nearby -System displays “No nearby alternatives found”. 2.If GPS data missing -System asks for manual location input. 3.If dataset incomplete -System marks missing dishes as “Data unavailable”. 4.If system response > 5s -System displays a loading message until it is ready.
Includes:	-Retrieving dish prices includes detecting categories with high inflation. -Searching dishes includes recommending alternative hawker meals. -Recommending cheaper alternatives includes showing estimated savings.
Special Requirements:	Location accuracy must be within 300 m.
Assumptions:	The user has internet access.

Notes and Issues:	GPS signals may be inaccurate indoors, affecting nearby stall suggestions

Use Case ID:	Use Case 5		
Use Case Name:	Grocery Basket Planner		
Created By:	Li Shuhui	Last Updated By:	Li Shuhui
Date Created:	2025.9.19	Date Last Updated:	2025.9.19

Actor:	-User (Student, Young Working Adult, Household) -System
Description:	The Grocery Basket Optimizer allows users to enter a shopping list and receive optimized basket meeting constraints.
Preconditions:	-User has logged into the website. -User has a shopping list ready to enter.
Postconditions:	User leaves the current page or switches to another module of the system.
Priority:	High
Frequency of Use:	Frequently.
Flow of Events:	1.User searches and selects meals from a predefined dataset. 2.User specifies the number of people to serve. 3.System generates a complete grocery shopping list with food items and required quantities.
Alternative Flows:	If user wants to download the shopping list -System generates the shopping list into a picture or a file, and also provides the tab to copy the list.
Exceptions:	1.If some items have no price data -System shows "Data not available,please check in person." 2.If system response > 5s -System displays a loading message until it is ready.
Includes:	-Searching and selecting meals from the dataset includes specifying the number of people served and restricting selection to available meals. -Generating a complete grocery shopping list includes showing estimated costs.

Special Requirements:	-The system shall restrict user input to only meals available in the dataset.
Assumptions:	The user has internet access.
Notes and Issues:	-Cost calculations are based on average prices; actual store prices may vary.